MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG5SECCND300 FOOD WASTE AND BYPRODUCT MANAGEMENT

Duration: 1 hr Maximum Marks: 35

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Each question is tagged with the appropriate Bloom's Revised Taxonomy level (Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.) and the relevant course outcome (CO) number.

* Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S).

Part A

Objective Questions

Answer **All** Questions

Each question carries **1** mark

1. Waste generated during food processing may include…….. [U][1]
   a) Peels and seeds b) Bones and shells c) Pulp and pomace d) All of the above
2. Which of the following is an example of solid waste generated by the fruit processing industry? [U][1]
   a) Fruit peels b) Wastewater c) Smoke d) Steam
3. BOD stands for: [U][1]
   a) Biological Oxygen Demand b) Biochemical Oxygen Demand c) Biological Organic Decomposition d) Biological Oxidation Demand.
4. Which of the following is a major source of waste in fruit processing? [U][2]
   a) Peels b) Seeds c) Pulp residues d) All of the above
5. Which of the following is a common by-product of rice milling? [U][2]
   a) Whey b) Rice bran c) Fish meal d) Pomace

[1 x 5 = 5 marks]

Part B

Short Answer Type Questions

Answer any **5** Questions

Each question carries **2** marks

6. Describe COD. [U][1]
7. Write on BOD. [C][1]
8. Explain the significance of temperature and pH in waste characterization. [U][1]
9. Discuss on any four major types of waste generated by food processing industries. [U][1]
10. Discuss any two uses of poultry waste. [U][2]
11. Write the major sources of waste in fruit processing? [C][2]

12. Discuss on any two methods of tapioca waste utilization. [U][2]

[2 x 5 = 10 marks]

Part C

Essay Type Questions

Answer **2** Questions

Each question carries **10** marks

13. Discuss the scope and importance of food waste management. [U][1]
14. Explain on types of waste. [U][1]
15. Explain the utilization of fruit waste for the production of vinegar and organic acids. [U][2]
16. Write on various applications of green leafy vegetable waste. [C][2]

[10 x 2 =20 marks]